

Scalability & Future Plan

Date	05 July 2026
Team ID	xxxxxxx
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Local CPU Inference Stalls: Model evaluation (BiLSTM + BERT) runs locally on a single host CPU thread.	Concurrent user volume causes immediate thread contention, spiking response latency past 3 seconds.	High
2	Volatile Local Logging Architecture: Interaction tracking logs are handled via a local standalone CSV framework.	Cannot scale to persistent distributed environments or cross-session user sessions without write conflicts.	High
3	Synchronous Third-Party API Dependency: Generative coping text relies on real-time synchronous Gemini API external network hops.	Frontend interface operations block completely until network generation packages resolve.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single host local development environment.	Migrate neural ciphers onto cloud server infrastructures (such as AWS EC2 or ECS containers) managed behind an Application Load Balancer with auto-scaling triggers.
2	Data Storage	Disconnected standalone execution log files.	Transition data pipelines into a scalable database system like MongoDB or PostgreSQL Cloud Service layers to store long-term interaction history seamlessly.

3	Inference Performance	Synchronous CPU matrix processing blocks.	Decouple heavy neural operations into an asynchronous processing worker queue utilizing Celery and Redis, shifting model hosting onto dedicated Triton Inference Servers or AWS SageMaker instances.
4	Security	Basic system environment parameters.	Implement OAuth2 token validation schemas, secure API gateway rate-limiting, and TLS/SSL transit encryption layers to protect private learner interaction telemetry.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Multi-Modal Input Integration (Voice-to-Text Speech Processing & Audio Sentiment Ciphers).	2 Months	Enables users to verbally communicate academic stress, improving dashboard accessibility and capturing structural audio-emotional signals.
Phase 3	Adaptive Learning Vector Pathing & Automatic Resource Recommendation.	4 Months	Recommends specific study links, target documentation, or relevant practice problems mapping directly to the subject where frustration was identified.
Phase 4	Cross-Platform Mobile Application Development & Advanced Departmental Analytics Portals.	6 Months	Expands user accessibility, tracks historical stress curves, and provides macro-level insight into widespread student curriculum bottlenecks.